THE NOVA RESEARCH CURRICULUM

General Science

Grade 6 · Term 3

Student edition

Outcomes: SC-6.7 series and parallel circuits, magnetism and electromagnets; SC-6.8 planets, dwarf planets, comets and satellites, including GPS.

Lesson notes

- **SC-6.7 Circuits (Weeks 1–4):** a circuit needs a cell, wires and a load in a complete loop; circuit symbols. Series: one path, bulbs share the push and are dimmer; if one breaks, all go out. Parallel: several paths, each bulb as bright; one can fail and the others stay on — that is how houses are wired. Conductors and insulators. Home lab: cell, bulb and foil wires. Never touch mains electricity.

  ▶ **Try it on NovaSchool:** Circuit builder: series and parallel — open **Simulations** on your dashboard.

- **SC-6.7 Magnets (Weeks 5–8):** north and south poles; like poles repel, unlike attract; magnetic materials (iron, nickel, cobalt). Magnetic field lines with iron filings. Electromagnet: coil of wire round an iron nail with a cell; more turns or more current make it stronger; it can be switched off. Uses: cranes in scrapyards, doorbells, motors.
- **SC-6.8 Space (Weeks 9–12):** eight planets in order; dwarf planets (Pluto); comets (ice and dust, tails pointing away from the Sun); natural satellites (the Moon) and artificial satellites. GPS uses signals from several satellites to find a position. Pakistan's satellites (e.g. PakSat-MM1).

  ▶ **Try it on NovaSchool:** Planets orbiting the Sun — open **Simulations** on your dashboard.

Workbook pages

Page	Task
1–2	Draw series and parallel circuits with symbols; predict what happens.
3	Electromagnet test: turns vs paper clips picked up (table and bar chart).
4	Solar System diagram in order.
5	How GPS finds you: short explanation with diagram.

Question bank

#	Outcome	Question	Your answer
1	SC-6.7	In a series circuit, if one bulb breaks …	the others get brighter / all go out / nothing changes / the cell charges
2	SC-6.7	Houses are wired in …	series / parallel / no circuit / a single loop
3	SC-6.7	Which is an insulator?	copper / iron / plastic / aluminium

#	Outcome	Question	Your answer
4	SC-6.7	Like poles of magnets attract. True or false.	True / False
5	SC-6.7	Name one way to make an electromagnet stronger.	_____
6	SC-6.7	Which is a magnetic material?	wood / iron / glass / copper
7	SC-6.8	Which planet is closest to the Sun?	Venus / Earth / Mercury / Mars
8	SC-6.8	Pluto is now called a …	star / comet / dwarf planet / moon
9	SC-6.8	The Moon is a natural satellite of the Earth. True or false.	True / False
10	SC-6.8	GPS finds your position using signals from …	one radio tower / several satellites / the Moon / a compass